



# ADITYA UNIVERSITY

**SCHOOL OF ENGINEERING**

**Department of DATA SCIENCE**

## **CERTIFICATE**

This is to certify that the Cornerstone project work entitled “**BUS BOOKING SYSTEM**” is submitted by

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in partial fulfillment of the requirements for the award of the B.Tech. degree in Data Science for the academic year 2025-2026.

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## DECLARATION

We hereby declare that the Cornerstone project entitled “**BUS BOOKING SYSTEM**” is original and is submitted to **ADITYA UNIVERSITY**, Surampalem, in partial fulfillment of the requirements for the award of the **B.Tech. degree** in **DATA SCIENCE**. We further declare that this project work has not been submitted, either in full or in part, for the award of any degree in this or any other institution. We also confirm that this work complies with the Institutional Academic Integrity and AI Usage Policy

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## **ABSTRACT**

The Bus Booking System is a web-based application developed to simplify, automate, and enhance the process of booking bus tickets online. In the present digital era, passengers prefer quick and convenient reservation systems, and this application provides an efficient platform for users to search available buses, view details such as timings, routes, and seat availability, and book tickets easily through an interactive and user-friendly interface. The system also allows users to manage their bookings and provides a centralized platform for handling reservations efficiently. It is developed using Java technologies, where Java Swing is used for designing the frontend interface, and JDBC and Servlets are used for backend processing, with MySQL serving as the database for storing and managing user and booking information. The system ensures secure user authentication and reliable data handling throughout the process. To improve reliability and reduce issues such as fake bookings or misuse, the system can include validation mechanisms such as booking confirmation and user verification. By automating the entire reservation process, the system reduces manual effort, minimizes errors, improves accuracy, and enhances overall efficiency. Additionally, it increases transparency, provides better user experience, and serves as an effective solution for modern bus transportation and ticket booking services.

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# ACRONYMS

- BBS : Bus Booking System
- JDBC : Java Database Connectivity
- GUI : Graphical User Interface
- SQL : Structured Query Language
- DB : Database
- UI : User Interface
- UX : User Experience
- API : Application Programming Interface
- JVM : Java Virtual Machine
- JDK : Java Development Kit
- JRE : Java Runtime Environment
- HTTP : HyperText Transfer Protocol
- HTTPS : HyperText Transfer Protocol Secure
- CRUD : Create, Read, Update, Delete

# **1. INTRODUCTION**

## **1.1 Brief Information about the Project**

The Bus Booking System (BBS) is a web-based application developed to manage bus ticket reservations efficiently. It provides a centralized platform for users to search available buses, view details such as routes, timings, and seat availability, and book tickets in a structured and convenient manner. The system replaces traditional manual ticket booking methods with a computerized solution, thereby improving accuracy, reducing time consumption, and enhancing user convenience. The application is developed using Core Java for programming, Servlets and JSP for handling web-based operations, JDBC for database connectivity, and MySQL database for storing user and booking information. It is designed to be user-friendly and suitable for use by passengers, travel agencies, and transportation service providers.

## **1.2 Motivation and Contribution of Project**

In many transportation systems, bus ticket booking is still done manually or through basic methods, which leads to inefficiencies such as difficulty in checking seat availability, time-consuming reservation processes, and increased chances of errors or overbooking. These limitations motivated the development of the Bus Booking System.

The key contributions of this project include:

- Providing a digital solution for booking bus tickets
- Reducing manual effort and paperwork
- Improving accuracy in seat allocation and booking management
- Enabling quick search and selection of buses based on routes and timings
- Enhancing user convenience through an easy-to-use interface

This system simplifies the bus reservation process and improves overall efficiency and reliability in transportation services.

### **1.3 Objectives of the Project**

The main objectives of the Bus Booking System are:

- To develop a user-friendly application for booking bus tickets online
- To store user and booking information in a structured database
- To provide functionalities for searching buses based on routes and timings
- To enable users to select seats and book tickets efficiently
- To display bus and booking details in an organized and clear format

### **1.4 Scope of the Project**

The Bus Booking System can be used by passengers, travel agencies, and transportation service providers to manage bus ticket reservations efficiently. It supports operations such as searching available buses, viewing routes and timings, selecting seats, booking tickets, and managing booking details. The system provides a simple and user-friendly interface for handling reservations and maintaining data in a centralized database. It can be used for both small and large-scale transportation services, improving convenience, accuracy, and overall efficiency in the booking process.

## 2. SYSTEM ANALYSIS

### 2.1 Functional Requirements

Functional requirements define the core operations and functionalities that the Bus Booking System (BBS) must perform to ensure efficient and reliable ticket booking and management.

The functional requirements of the Bus Booking System are:

- **User Registration and Login**

The system should allow users to register and log in securely using their credentials. Proper validation should be applied to ensure data accuracy and security.

- **Search Buses**

The system should allow users to search for buses based on source, destination, and journey date. It should display available buses with relevant details.

- **View Bus Details**

The system should display detailed information such as bus name, departure time, arrival time, seat availability, and ticket price..

- **Select Seats**

The system should allow users to select available seats visually. It should prevent multiple users from booking the same seat.

- **Book Ticket**

The system should allow users to book tickets after selecting the bus and seats. A confirmation message should be provided after successful booking.

- **View Booking Details**

The system should allow users to view their booking summary, including journey details, selected seats, and total fare.

- **Booking History**

The system should store and display previous bookings of the user for future reference.

- **Admin Management**

The system should allow administrators to manage bus details, routes, schedules, and pricing efficiently.

## 2.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and performance standards that the Bus Booking System must satisfy.

- **Performance**

The system should respond quickly to operations such as searching buses, selecting seats, and booking tickets without noticeable delays.

- **Security**

The system should ensure secure user authentication and protect user data. Only authorized users should be allowed to access and perform booking operations.

- **Usability**

The system should be simple, intuitive, and user-friendly, allowing users to easily search buses, select seats, and complete bookings without difficulty.

- **Reliability**

The system should function consistently without failures and should handle errors properly without loss of booking or user data.

- **Scalability**

The system should be capable of handling an increasing number of users, bookings, and bus data without affecting performance.

## 2.3 Requirements Specification

This section specifies the hardware and software needed to run the system.

### 2.3.1 Minimum Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4 GB (minimum)
- Hard Disk: 20 GB free space
- Monitor: Standard display
- Keyboard and Mouse

### ***2.3.2 Software Requirements***

- **Operating System:** Windows 7/10/11
- **Programming Language:** Java
- **IDE:** VS Code
- **Database:** Oracle Database
- **Connectivity:** JDBC

## **3. TECHNOLOGY DESCRIPTION**

### **3.1 Programming Language**

#### **3.1.1 Introduction to Java**

Java is a high-level, object-oriented programming language developed by Sun Microsystems. It is widely used for developing desktop, web, and enterprise applications due to its platform independence and robustness.

In the Bus Booking System, Java is used to implement the core functionality of the application. It handles operations such as searching buses, managing bookings, and retrieving booking details. Java provides features like object-oriented programming, exception handling, and strong memory management, which help in building reliable and efficient applications.

One of the main advantages of Java is that it is platform-independent, meaning the program can run on any system that has a Java Virtual Machine (JVM). This makes it suitable for developing portable applications like the Bus Booking System.

### **3.2 Framework / Libraries**

#### **3.2.1 Introduction to Swing**

Swing is a part of Java that is used to create graphical user interfaces (GUI). It provides a set of components such as buttons, labels, text fields, tables, and frames to design user-friendly applications.

In this project, Swing is used to design the user interface of the Bus Booking System. It allows users to interact with the system easily through windows such as:

- Login Window
- Bus Search Window
- Seat Selection Window
- Booking Page
- Booking Details / History Page

Swing provides flexibility in designing layouts and improves the overall user experience. It makes the application visually interactive and easy to use.

### **3.3 Database Technology**

The Bus Booking System uses a relational database for storing and managing bus, user, and booking information in a structured manner. The database ensures efficient data storage, retrieval, and manipulation, which is essential for maintaining

accuracy and consistency in the system. In this project, MySQL is used as the backend database, and JDBC is used for connecting the Java application to the database.

### **3.3.1 JDBC (Java Database Connectivity)**

JDBC is an API (Application Programming Interface) used in Java to connect and interact with databases. It allows Java applications to execute SQL queries and retrieve or update data in the database.

In the Bus Booking System, JDBC is used to connect the Java application with the MySQL database. It performs operations such as:

- Establishing a connection to the database
- Executing SQL queries (INSERT, UPDATE, DELETE, SELECT)
- Retrieving data from the database
- Closing the database connection

JDBC acts as a bridge between the application and the database, ensuring smooth communication and data transfer. It helps in maintaining data consistency and enables efficient data handling.

### **3.3.2 MySQL Database**

MySQL is a widely used relational database management system used for storing and managing structured data. It provides high performance, reliability, and security for handling large volumes of data.

In the Bus Booking System, MySQL is used to store data such as user details, bus information, routes, bookings, and seat availability. It supports SQL for data manipulation and ensures data integrity through constraints like primary keys and data types.

## **3.4 Tools & IDEs**

### **3.4.1 Visual Studio Code**

Visual Studio Code (VS Code) is used as the development environment for the Bus Booking System. It is a lightweight and powerful code editor that supports Java programming through extensions. VS Code provides features such as syntax highlighting, code completion, and debugging support, which help improve coding efficiency. Its simple and user-friendly interface makes it suitable for developing and testing the application.

## 4. DATABASE DESIGN

### 4.1 Database Tables

The Bus Booking System uses a relational database to store data in a structured manner.

The database consists of multiple tables such as User, Bus, Booking, and Route.

- The **User** table stores details of users such as user ID, name, email, and contact information.
- The **Bus** table stores information about buses including bus ID, bus name, timings, and seat capacity.
- The **Route** table stores details of source and destination locations along with route information.
- The **Booking** table stores booking details such as booking ID, user ID, bus ID, seat numbers, and travel date.

This separation of data into different tables helps in reducing redundancy, maintaining data consistency, and improving overall database organization.

### 4.2 Table Structure

#### 4.2.1 Users Table

| Field Name | Data Type    | Description   | Constraint  |
|------------|--------------|---------------|-------------|
| id         | INT          | User ID       | Primary Key |
| name       | VARCHAR(100) | User Name     | Not Null    |
| email      | VARCHAR(100) | Email Address | Unique      |
| password   | VARCHAR(100) | Password      | Not Null    |

#### 4.2.2 Buses Table:

| Field Name     | Data Type    | Description    | Constraint  |
|----------------|--------------|----------------|-------------|
| id             | INT          | Bus ID         | Primary Key |
| bus_name       | VARCHAR(100) | Bus Name       | Not Null    |
| departure_time | VARCHAR(10)  | Departure Time | Not Null    |
| arrival_time   | VARCHAR(10)  | Arrival Time   | Not Null    |
| bus_type       | VARCHAR(50)  | Bus Type       | -           |
| seats          | INT          | Total Seats    | Not Null    |

### 4.2.3 Bookings Table

| Field Name   | Data Type    | Description    | Constraint  |
|--------------|--------------|----------------|-------------|
| id           | INT          | Booking ID     | Primary Key |
| username     | VARCHAR(100) | User Name      | Not Null    |
| from_city    | VARCHAR(50)  | Source         | Not Null    |
| to_city      | VARCHAR(50)  | Destination    | Not Null    |
| journey_date | DATE         | Travel Date    | Not Null    |
| seats        | VARCHAR(50)  | Selected Seats | Not Null    |
| price        | DOUBLE       | Total Price    | Not Null    |

## 4.3 Keys and Constraints

Keys and constraints are used to maintain data integrity and ensure accurate data storage in the Bus Booking System.

- **Primary Keys:**

- o id in Users table
- o id in Buses table
- o id in Bookings table

- **Foreign Key:**

- o username in Bookings table references Users table
- o bus\_id (if used) in Bookings table references Buses table

- **NOT NULL Constraint:**

- o Important fields such as user name, email, bus name, source, destination, and journey date cannot be left empty

- **Data Type Constraints:**

- o Each field accepts only valid data types such as INT, VARCHAR, DATE, and DOUBLE to ensure correctness

These constraints help in maintaining data consistency, avoiding duplicate entries, and ensuring reliable data management.

## 4.4 Table Relationships

The tables in the Bus Booking System are related through fields such as user details and booking information.

- Each booking is associated with one user
- One user can have multiple bookings
- Each booking is associated with one bus
- One bus can have multiple bookings

This represents a **one-to-many relationship** between Users and Bookings, and also between Buses and Bookings.

## 5. SYSTEM DESIGN

### 5.1 System Architecture

The Bus Booking System follows a three-layer architecture:

1. **Presentation Layer (User Interface)**
  - o Developed using JSP and HTML/CSS
  - o Allows users to interact with the system through pages like login, bus search, seat selection, and booking
2. **Application Layer (Business Logic)**
  - o Developed using Java Servlets
  - o Processes user inputs and performs operations such as user authentication, bus search, seat selection, and booking management
3. **Data Layer (Database)**
  - o Stores user details, bus information, and booking data
  - o Managed using MySQL database through JDBC

This architecture ensures better organization, security, and maintainability of the system.

### 5.2 Modules Description

The system is divided into the following modules:

1. **User Authentication Module**
  - Allows users to register and log in securely
  - Validates user credentials
  - Maintains user session
2. **Bus Search Module**
  - Allows users to search buses based on source, destination, and date
  - Displays available buses with details like timings and price
3. **Seat Selection Module**
  - Allows users to select available seats
  - Prevents double booking of seats
  - Displays seat availability status
4. **Booking Module**
  - Handles ticket booking process

- Stores booking details in the database
- Calculates total price based on selected seats

5. **Booking Management Module**

- Displays booking summary and confirmation
- Maintains booking history of users

6. **Database Connectivity Module**

- Connects Java application with MySQL database using JDBC
- Executes SQL queries for data operations

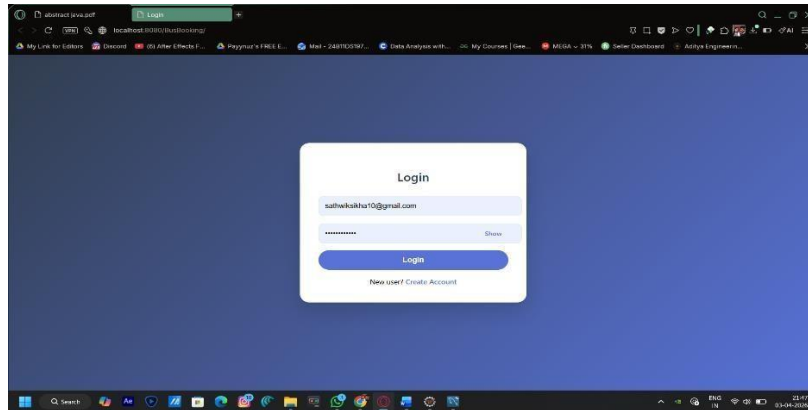
7. **User Interface Module**

- Provides web-based interface using JSP, HTML, and CSS
- Handles user interactions and navigation

## 6. SYSTEM IMPLEMENTATION

### Login Screen:

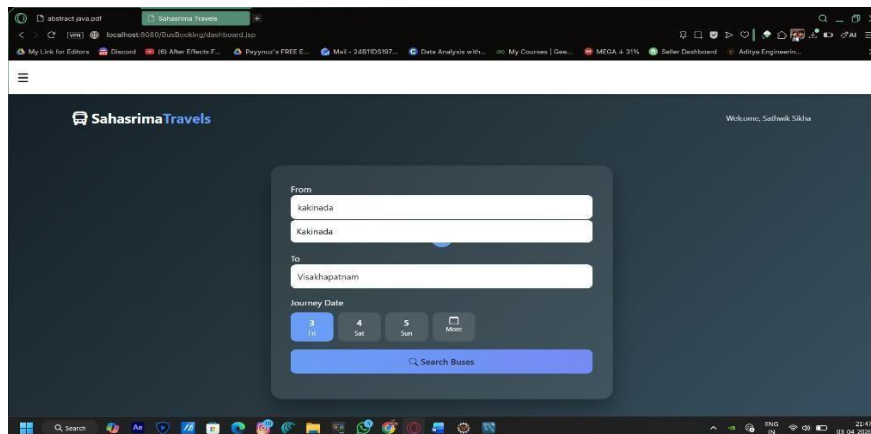
Displays the login interface where the user enters email and password to access the system. It provides secure authentication and allows only registered users to log in.



**Figure 1: Login Screen of Bus Booking System**

### Dashboard Screen:

Displays the main interface where users can search for buses by entering source, destination, and journey date. It provides easy navigation for booking tickets.



**Figure 2: Dashboard Screen of Bus Booking System**

### Search Results Screen:

Displays the list of available buses based on user input. It shows details such as bus name, timings, seat availability, and ticket price.

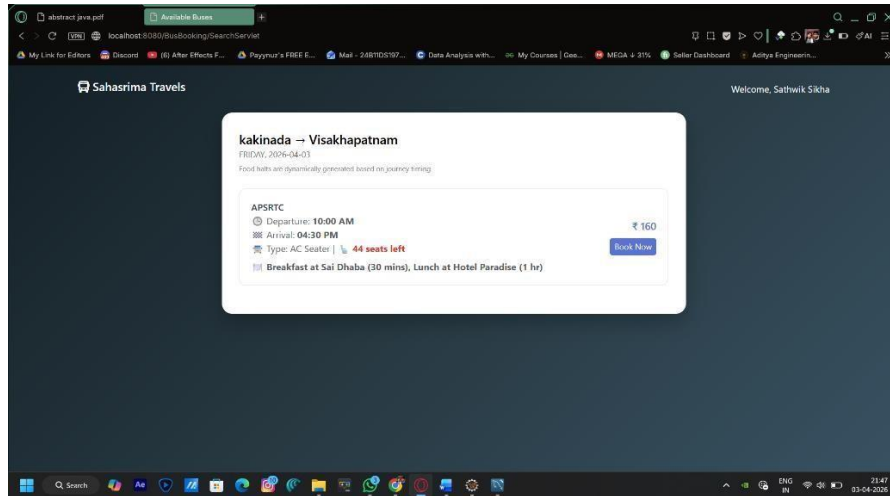


Figure 3: Search Results Screen

### Seat Selection Screen:

Allows users to select available seats visually. Different colors indicate available, selected, and booked seats to avoid conflicts.

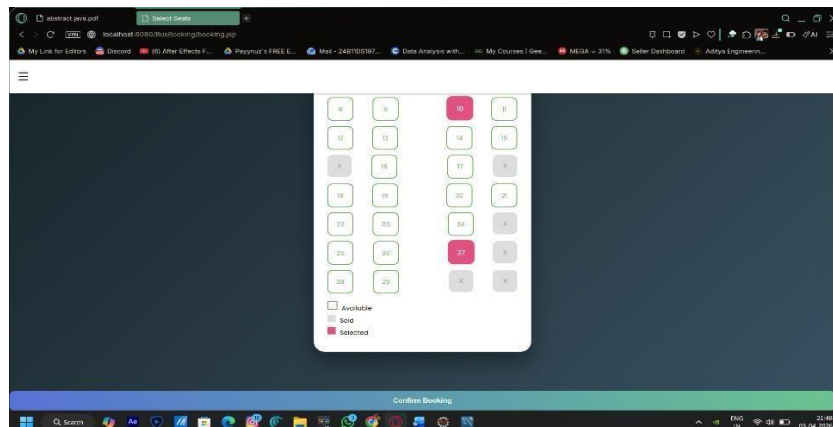
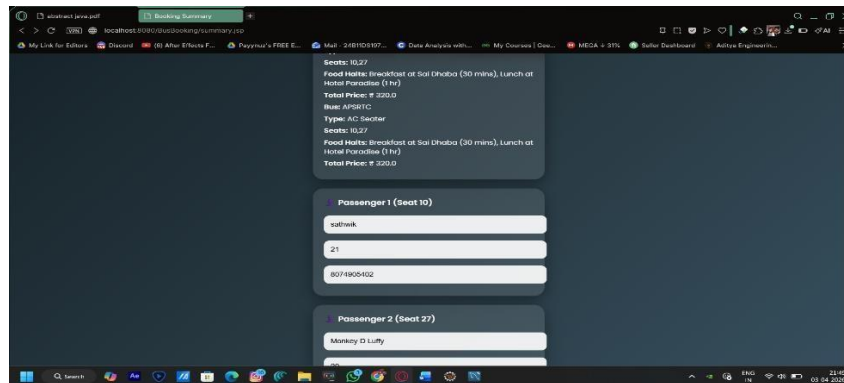


Figure 4: Seat Selection Screen

## Booking Summary Screen

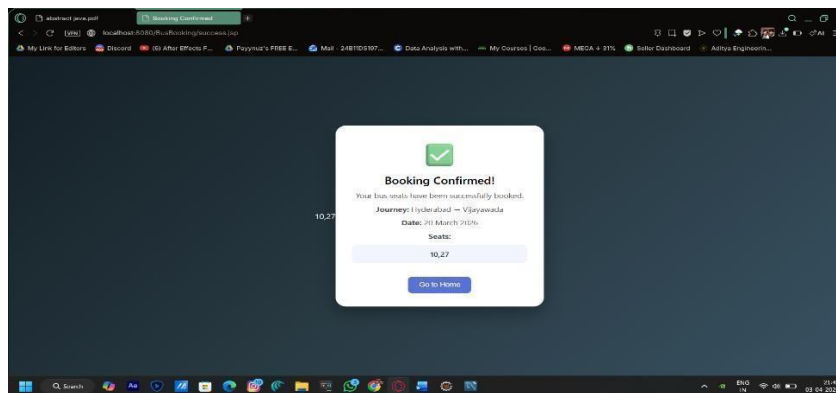
Displays booking details including selected seats, passenger information, and total price before confirmation.



**Figure 5: Booking Summary Screen**

## Booking Confirmation Screen

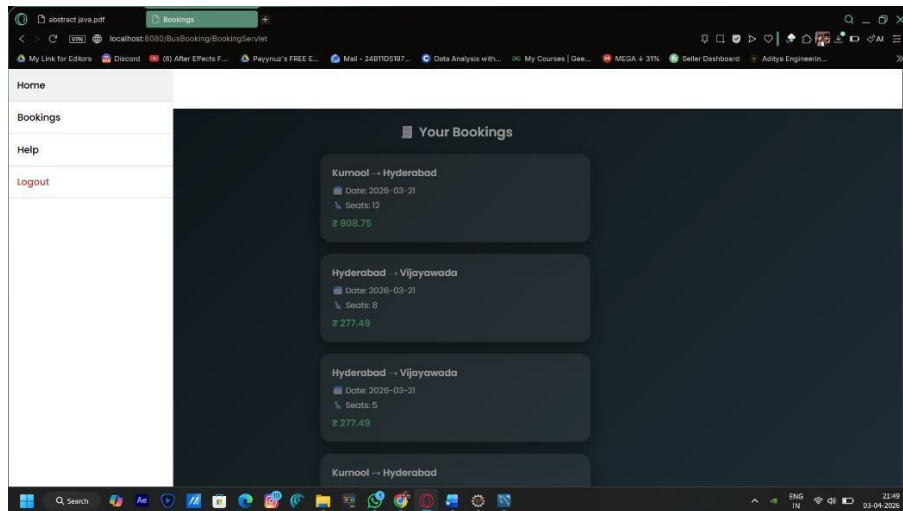
Shows confirmation message after successful booking with details like journey, date, and seat numbers.



**Figure 6: Booking Confirmation Screen**

## My Bookings Screen

Displays all previous bookings made by the user along with details such as journey route, date, and total fare.



**Figure 7: My Bookings Screen**

## **7. CONCLUSION**

The Bus Booking System has been successfully developed to manage bus ticket reservations in an efficient and organized manner. The system replaces traditional manual booking methods with a computerized solution, thereby reducing errors, saving time, and improving overall accuracy.

The application provides essential functionalities such as user registration and login, searching buses, selecting seats, booking tickets, and viewing booking details through a user-friendly interface. By using technologies like Java, JSP, Servlets, JDBC, and MySQL database, the system ensures reliable performance, secure data handling, and smooth interaction between different components.

Overall, the system improves the efficiency of the booking process and enhances user convenience. It reduces manual workload and provides a scalable solution that can be further enhanced in the future by adding features such as online payment integration, real-time tracking, and advanced booking management.

In conclusion, the Bus Booking System meets the project objectives and serves as an effective solution for modern bus ticket reservation systems.

## 8. FUTURE ENHANCEMENTS

The Bus Booking System developed in this project provides basic functionalities for managing bus ticket reservations. However, the system can be further enhanced by adding advanced features to improve its functionality and usability.

Some possible future enhancements include:

- **Online Payment Integration:**

Adding secure payment options such as UPI, credit/debit cards, and digital wallets

- **Real-Time Bus Tracking:**

Integrating GPS-based tracking to allow users to monitor the live location of buses

- **Seat Availability in Real-Time:**

Providing live updates of seat availability to avoid booking conflicts

- **Multi-User Support:**

Allowing multiple users to access and use the system simultaneously

- **Improved User Interface:**

Enhancing the design and responsiveness for a better user experience

- **Notification System:**

Sending booking confirmations and updates through SMS or email

These enhancements can make the system more scalable, secure, and suitable for real-time transportation and booking services.

## 9. REFERENCES

The following references were used during the development of the Bus Booking System:

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## APPENDIX

### A.1 Sample Code

#### Database Connection Code:

The following code establishes a connection between the Java application and the MySQL database using JDBC.

#### DBConnection.java:

```
import java.sql.*;

public class DBConnection {
    public static Connection getConnection() {
        Connection con = null;
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/bus_booking", "root",
                "password");
        } catch (Exception e) {
            System.out.println(e);
        }
        return con;
    }
}
```

#### Insert Booking Record:

This code is used to add a new booking into the database.

#### InsertBooking.java:

```
import java.sql.*;

public class InsertBooking {
    public static void main(String[] args) {
        try {
            Connection con = DBConnection.getConnection();

            String query = "INSERT INTO bookings VALUES(?,?,?,?,?,?,?)";
            PreparedStatement pst = con.prepareStatement(query);
```

```

        pst.setInt(1, 1);
        pst.setString(2, "Sathwik");
        pst.setString(3, "Hyderabad");
        pst.setString(4, "Vijayawada");
        pst.setDate(5, Date.valueOf("2026-04-10"));
        pst.setString(6, "A1,A2");
        pst.setDouble(7, 1200);

        int i = pst.executeUpdate();

        if (i > 0)
            System.out.println("Booking Added Successfully");

        con.close();
    } catch (Exception e) {
        System.out.println(e);
    }
}
}

```

## A.2 SQL Queries

### Create Users Table:

```

CREATE TABLE users (
    id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(100) UNIQUE,
    password VARCHAR(100) NOT NULL
);

```

**Create Buses Table:**

```
CREATE TABLE buses (  
    id INT PRIMARY KEY AUTO_INCREMENT,  
    bus_name VARCHAR(100) NOT NULL,  
    departure_time VARCHAR(10),  
    arrival_time VARCHAR(10),  
    bus_type VARCHAR(50),  
    seats INT  
);
```

**Create Bookings Table:**

```
CREATE TABLE bookings (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    username VARCHAR(100),  
    from_city VARCHAR(50),  
    to_city VARCHAR(50),  
    journey_date VARCHAR(50),  
    seats VARCHAR(50),  
    price DOUBLE,  
    passenger_names TEXT,  
    passenger_ages TEXT,  
    passenger_phones TEXT  
);
```